Tina4 Ruby – Database Class

🔥 **Hot Tips**

- Set `DATABASE_URL` in `.env` and Tina4 auto-connects on startup
- `Tina4.database` gives you the global database instance
- Most methods return a `DatabaseResult` which you can transform
- Use `transaction { }` blocks for safe multi-statement operations

Connection

ruby

```ruby
require "tina4"

# SQLite (file)
db = Tina4::Database.new("sqlite3:data.db")

# SQLite (in-memory – perfect for tests)
db = Tina4::Database.new("sqlite3::memory:")

# PostgreSQL
db = Tina4::Database.new("postgres://user:pass@localhost:5432/mydb")

# MySQL / MariaDB
db = Tina4::Database.new("mysql://root:secret@localhost:3306/mydb")

# Microsoft SQL Server
db = Tina4::Database.new("mssql://sa:Password@localhost:1433/mydb")
```

```ruby
# Firebird
db = Tina4::Database.new("firebird://sysdba:masterkey@localhost:3050/path/db.fd
```

Required Gems

Driver	Gem	Install
SQLite	`sqlite3`	`gem install sqlite3`
PostgreSQL	`pg`	`gem install pg`
MySQL/MariaDB	`mysql2`	`gem install mysql2`
SQL Server	`tiny_tds`	`gem install tiny_tds`
Firebird	`fb`	`gem install fb`

Auto-Connection via `.env`

env

```
DATABASE_URL=sqlite3:data.db
```

ruby

```ruby
# After Tina4.initialize!, use:
db = Tina4.database
```

Core Methods

Method	Description	Returns
`execute(sql, params)`	Run any SQL (CREATE, DROP, INSERT, ...)	—
`insert(table, data)`	Smart insert (returns last_id)	`{ success: true, last_id: N }`

Method	Description	Returns
`update(table, data, filter)`	Update by filter hash	`{ success: true }`
`delete(table, filter)`	Delete by filter hash	`{ success: true }`
`fetch(sql, params, limit:, skip:)`	SELECT with pagination	`DatabaseResult`
`fetch_one(sql, params)`	Return single row as hash	`Hash` or `nil`
`table_exists?(name)`	Check if table exists	`bool`
`tables`	List all tables	`Array`
`columns(table)`	Get column info	`Array`
`transaction { }`	Block-based transaction	—
`close`	Close connection	—

Result Object

Every `fetch` returns a `DatabaseResult` with these methods:

ruby

```ruby
result = db.fetch("SELECT * FROM users")

result.to_a          # → Array of hashes
result.count         # → Integer
result.first         # → First row hash
result.to_json       # → JSON string
result.to_csv        # → CSV string
result.columns       # → Column names
result.empty?        # → Boolean
```

Examples

ruby

```ruby
db = Tina4::Database.new("sqlite3::memory:")

# Create table
db.execute("CREATE TABLE users (id INTEGER PRIMARY KEY, name TEXT, age INTEGER)

# Insert
db.insert("users", { name: "Alice", age: 30 })
db.insert("users", { name: "Bob", age: 25 })

# Update
db.update("users", { age: 31 }, { id: 1 })

# Delete
db.delete("users", { id: 2 })

# Select
result = db.fetch("SELECT * FROM users")
puts result.to_a     # → [{ id: 1, name: "Alice", age: 31 }]
puts result.count    # → 1

# With parameters
result = db.fetch("SELECT * FROM users WHERE age > ?", [30])
puts result.first[:name]  # → "Alice"

# Pagination
result = db.fetch("SELECT * FROM users", [], limit: 10, skip: 20)

db.close
```

Transactions

ruby

```ruby
db.transaction do |tx|
  tx.insert("users", { name: "Risky" })
  tx.insert("users", { name: "Also Risky" })
  # Auto-commits on success, auto-rollbacks on exception
end
```

One-liner Heaven (Tina4 style)

ruby

```ruby
Tina4::Database.new("sqlite3:data.db").execute("INSERT INTO logs (msg) VALUES (
```

That's it. No models. No config files. No nonsense.

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